

GUARDSHIELD AI — TEAMMATE PROJECT GUIDE

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Project Mentor: Dr. Abhishek Kaushal | **Project Domain:** Generative AI & AI Safety (MLOps)

Hi Rachit! This document explains your exact role on our project in simple, everyday English. You handled the **Academic Research and Theoretical Foundation** (literature review of 6 top benchmark papers, discovering the 3 core research gaps, and formulating the math behind token entropy and NLI verification). Use this guide to understand what you researched, why our project is scientifically novel, and how to comfortably present it to our mentor.

1. What Was Your Exact Role in the Project? (The Big Picture)

Before any serious AI project writes a single line of code, you have to answer two critical academic questions: *'What did previous scientists do?'* and *'Where did they fail that our project solves?'*

Your responsibility on the team was the Scientific & Research Core:

- **Literature Survey:** Thoroughly analyzing 6 major research papers from top venues (Cambridge, Meta AI, NeurIPS, EMNLP, ACL).
- **Research Gap Formulation:** Discovering that existing tools suffer from a 3-second offline lag and 8-Billion parameter bloat.
- **Novel Solution Formulation:** Formulating our 3 original research contributions (Mid-Sentence Early Termination, 99% parameter reduction, Hinglish defense).
- **Algorithmic & Mathematical Modeling:** Defining Shannon Token Logit Entropy $H(x)$ and Windowed Cross-Encoder NLI logic.
- **Paper Publication Plan:** Structuring the IEEE conference research paper draft for semester submission.

2. The 6 Major Research Papers You Surveyed (In Plain English)

You studied how tech giants and top universities currently attempt AI safety, identifying their critical flaws:

Paper Title & Authors	Conference / Year	What They Did	The Big Flaw You Found (The Gap!)
SelfCheckGPT (Manakul et al., Cambridge)	EMNLP 2023	Sampled multiple LLM responses to detect hallucinations without external databases.	Heavy Offline Delay: Takes 3 to 5 seconds AFTER response finishes. Unusable for live streaming!
Llama Guard (Inan et al., Meta AI)	Meta AI 2024	Fine-tuned an 8-Billion parameter Llama model to classify input/output safety.	Too Heavy & Costly: An 8B model requires expensive cloud GPUs. Cannot run as a lightweight sidecar.
JailbreakBench (Chao et al.)	NeurIPS 2024	Standardized adversarial jailbreaks across 10 safety categories.	Focused only on offline prompt evaluation, not live token-level streaming defense.
HaluEval (Li et al.)	EMNLP 2023	Built 35K hallucination examples across dialogue, QA, and summarization.	Purely an evaluation benchmark; provides no real-time inline mitigation mechanism.
CAPTURE Benchmark (Zhang et al.)	ACL 2025	Evaluated context-aware multi-lingual prompt injections.	Proved that commercial APIs fail on non-English code-mixed and Base64 attacks.
TruthfulQA (Lin et al.)	ACL 2022	Benchmarked how LLMs mimic popular human falsehoods and superstitions.	Demonstrated that larger models actually hallucinate falsehoods more confidently.

3. The 3 Original Research Gaps You Formulated (Our Scientific Novelty)

Mentors love clear research gaps! You formulated 3 novel contributions that give GuardShield AI high academic standing:

Contribution 1: Mid-Sentence Early Token-Termination (<25ms)

The Problem: Cambridge's SelfCheckGPT forces users to wait 3–5 seconds after the AI stops talking to verify facts.

Your Solution: We monitor token probability uncertainty in real time. The moment the AI begins hallucinating mid-sentence, our proxy detects the entropy spike and cuts off generation immediately with <25ms overhead.

Contribution 2: 99% Parameter Reduction vs. Meta's Llama Guard

The Problem: Meta uses an 8-Billion parameter model to check safety, which requires costly data center GPUs.
Your Solution: We fine-tuned an 86-Million parameter Small Language Model (DeBERTa-v3). It achieves high detection accuracy while running locally on a standard laptop at **■0 monetary cost**.

Contribution 3: Obfuscated & Multi-Lingual Attack Mitigation

The Problem: ACL 2025 research proved existing defenses fail against Base64 and Hinglish attacks.
Your Solution: We integrated a dual-stage architecture specifically evaluated against code-mixed and ciphered attacks.

4. The Math and Algorithms in Simple Everyday Language

Don't let the mathematical formulas intimidate you! Here is how to explain them simply to anyone:

- A. Token Logit Entropy $H(x) = - \sum p(i) \cdot \log[p(i)]$**
- *What it means:* Entropy measures 'uncertainty' or 'confusion'.
 - When an LLM is stating a well-known fact (e.g., '*The capital of France is Paris*'), the probability for 'Paris' is 99%. Entropy is near 0.
 - But when an LLM starts making up a fake medical claim or fake date, its probability gets scattered across dozens of random words. Entropy spikes above 1.20 bits! That sudden spike is our signal to investigate.
- B. Windowed Cross-Encoder NLI (Natural Language Inference)**
- *What it means:* Like a digital fact-checker.
 - If entropy spikes, we take the AI's sentence (Hypothesis) and compare it against the source document (Premise).
 - The Cross-Encoder outputs 3 scores: **Entailment** (Factually Supported), **Neutral**, or **Contradiction** (Hallucination!). If contradiction is high, output is cut off immediately.

5. Timeline: What You Have Completed Till Now (Week 1 to Week 7)

Timeline	What You Completed (Rachit's Contributions)	Research Deliverables
Weeks 1–2	Conducted literature survey of 6 benchmark papers; formulated research gap and project synopsis.	Amity Project Synopsis Draft
Week 3	Formulated mathematical foundation for token logit entropy calculation and sliding window size.	Mathematical Formulation Doc
Week 4	Designed loss function (cross-entropy with class weighting) and DeBERTa sequence classification head.	Classification Architecture Specs
Week 5	Benchmarked sub-25ms latency constraint; verified feasibility on local RTX 3050 CUDA GPU.	Latency Constraint Report
Weeks 6–7	Designed consecutive spike early termination logic and NLI entailment cross-attention loop.	src/nli_verifier.py specs

6. How to Present to Mentor Dr. Abhishek Kaushal (Your Spoken Script)

When the mentor asks: '*Rachit, explain the research foundation and novelty of your project*', confidently deliver this exact pitch:

"Good morning Sir. My primary role in GuardShield AI is the Research and Theoretical Formulation. Before building the system, I conducted a deep literature survey across 6 benchmark papers from Cambridge, Meta AI, NeurIPS, and EMNLP. Through this, I identified two critical scientific gaps: First, existing hallucination detectors like SelfCheckGPT operate strictly offline, adding a 3 to 5 second delay after the full text finishes generating. This destroys the real-time streaming experience of modern LLMs. Second, safety models like Meta's Llama Guard require massive 8-Billion parameter models that demand costly GPU infrastructure. To solve this, I formulated our dual-stage architecture: an 86-Million parameter DeBERTa-v3 Small Language Model that filters prompt injections in under 20ms (Pre-Scan), and an online token logit entropy monitoring algorithm that tracks token uncertainty mid-sentence (Post-Scan). When entropy spikes above 1.20 bits, we cross-verify with a windowed NLI Cross-Encoder to terminate hallucinations in real time. This achieves sub-25ms overhead and a 99% parameter reduction at zero hardware cost. We are currently drafting our findings for an IEEE conference paper."

7. Top 5 Viva / Guide Questions & Your Winning Answers

Q1: What is the exact research gap your project solves?

Answer: 'Sir, existing state-of-the-art like SelfCheckGPT takes 3 to 5 seconds to evaluate factuality because it samples multiple full responses offline. Our research gap addresses live streaming: we monitor token logit entropy mid-sentence, terminating hallucinations in under 25ms without breaking the SSE token stream.'

Q2: Why use an 86M parameter DeBERTa model instead of Meta's 8B Llama Guard?

Answer: 'Sir, Meta's Llama Guard has 8 Billion parameters, requiring expensive cloud GPU clusters. For production sidecar proxies, latency and cost are paramount. By fine-tuning an 86M DeBERTa-v3 SLM, we achieve a 99% parameter reduction and sub-20ms inference, running locally at \$0 cost.'

Q3: Explain the intuition behind Token Logit Entropy in simple words.

Answer: 'Sir, Shannon entropy measures prediction uncertainty. When an LLM generates factual text, the probability distribution is sharp and entropy is low. When the model hallucinates ungrounded facts, the probability spreads across multiple tokens, causing an entropy spike (>1.20 bits) that triggers our safety circuit breaker.'

Q4: What happens if the AI talks about a creative topic? Won't entropy always be high?

Answer: 'Sir, excellent question. That is why entropy is only a fast trigger, not the final judge! When an entropy spike occurs, we invoke a windowed Cross-Encoder NLI model to check if the statement contradicts reference context before terminating, preventing false alarms.'

Q5: Where do you plan to publish this research, and what is the paper status?

Answer: 'Sir, we are targeting an IEEE conference in AI Safety / Natural Language Processing (such as IEEE Confluence or IEEE INDICON). We have outlined the Abstract, Literature Review, and Methodology, and we will benchmark final F1 and latency metrics in Semester 7.'